

File Edit Selection View Go Run Terminal Help

EXPLORER

PYTHON_TEST

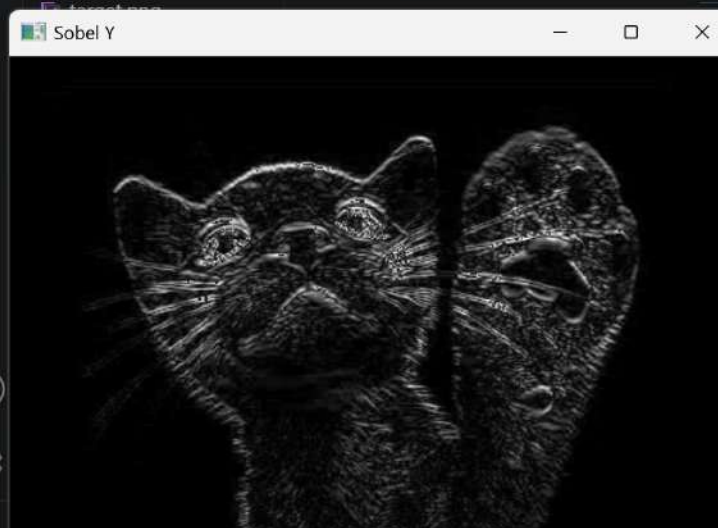
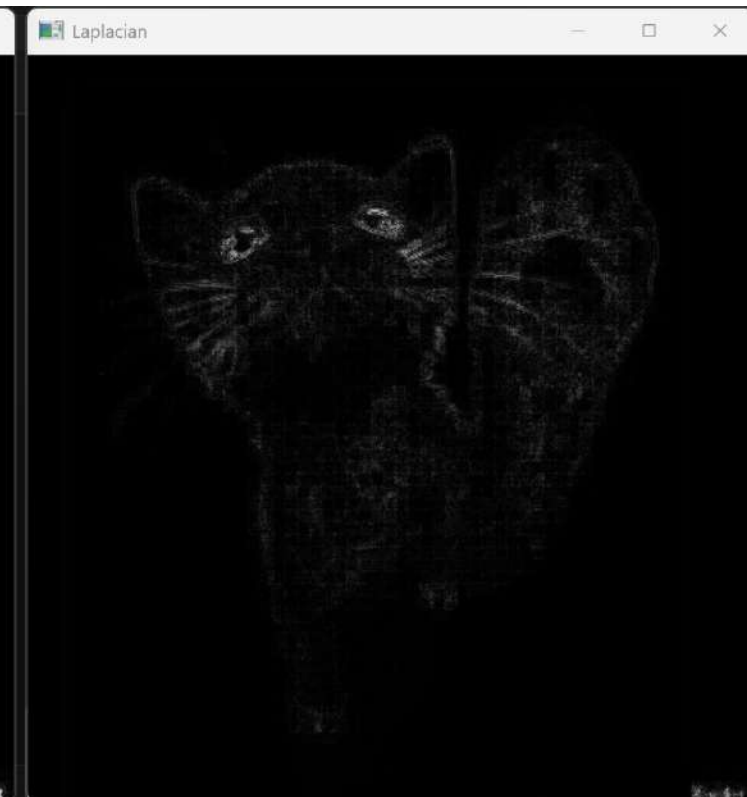
- masked.png
- my_masterpiece.png
- noise_reduction_pipel...
- otsu_and_riddler.py
- otsu_histogram.png
- output.bmp
- output.jpg
- output.tiff
- panorama_prep.png
- pixel_basics.py
- red.jpg
- region_comparison.png
- resize.py
- rotate.py
- saved_image.jpg
- saved_image.png
- simple_thresholding.py
- smart_crop.png
- sobel_and_laplacian.py

WELCOME

sobel_and_laplacian.py > ...

```
28 lap = cv2.Laplacian(image, cv2.CV_64F)
29 # Line 15: Take the absolute value of the Laplacian
30 # and convert it to an unsigned char
31 lap = np.uint8(np.absolute(lap))
32 # Line 16: Display the edge detection image
33 cv2.imshow("Laplacian", lap)
34 # Line 17: Wait indefinitely for a key press
35 # This prevents the windows from closing
36 cv2.waitKey(0)
37 # Line 18: Apply the Sobel operator
38 # The parameters (1, 0) indicate
39 # CV_64F allows capturing negative values
40 sobelX = cv2.Sobel(image, cv2.CV_64F, 1, 0)
41 # Line 19: Apply the Sobel operator
42 # The parameters (0, 1) indicate
43 sobelY = cv2.Sobel(image, cv2.CV_64F, 0, 1)
44 # Line 20: Blank line for reference
45 # Line 21: Convert Sobel X to unsigned char
46 # This makes the image suitable for display
47 sobelX = np.uint8(np.absolute(sobelX))
48 # Line 22: Convert Sobel Y to unsigned char
49 sobelY = np.uint8(np.absolute(sobelY))
```

PROBLEMS OUTPUT DEBUG CONSOLE



File Edit Selection View Go Run Terminal Help python_test

EXPLORER

- PYTHON_TEST
 - 2d_color_projections.p...
 - 2d_histograms.png
 - 3d_color_distribution....
 - 3d_histogram_projecti...
 - adaptive_thresholding...
 - annotated_image.jpg
 - annotated.png
 - art_cyanotype.png
 - art_green_screen.png
 - art_magenta_dream.p...
 - art_sepia.png
 - background_removed....
 - before_after.png
 - black.jpg
 - blended.png
 - blurring.py
 - canny.py
 - cat.png
 - challenge_blend.png
 - challenge_main.png
 - clock.png
 - color_histograms.png
 - color_histograms.py
 - color_isolation_blue.png
 - color_isolation_green....
 - color_isolation_purple....
 - color_isolation_red.png
 - color_isolation_yellow....
 - color_splash.png
 - composite_masterpiec...
 - contrast_enhancemen...
 - create_challenges_ima...
 - create_test_image_dav...
- OUTLINE
- TIMELINE

canopy.py

```
12
13 # Line 5: Add a command-line argument definition for input image
14 ap.add_argument(
15     # Line 6: Define short (-i) and long (--image) argument names
16     "-i", "--image",
17     # Line 7: Make this argument required when running the script
18     required=True,
19     # Line 8: Provide help message shown in terminal when using --help
20     help="Path to the image"
21 )
22 # Line 9: Close the add_argument() function call
23
24 # Line 10: Parse the command-line arguments into a dictionary
25 args = vars(ap.parse_args())
26
27 # Line 11: Load the image from the file path provided in arguments
28 image = cv2.imread(args["image"])
29
30 # Line 12: Convert the image from BGR color space to grayscale
31 image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
32
33 # Line 13: Apply Gaussian blur to reduce noise and smooth the image
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\raghu\OneDrive\Desktop\python_test> python canny.py --image cat.png
```

Ln 46, Col

